

Visualization & Dashboard Plan

Extending the EDA Findings Toward the Interactive Dashboard Deliverable

Data Visualization Design Project — May 2026

1. Purpose of This Document

The team report (Team_Report_EDA_Findings.docx) established WHAT drives customer satisfaction. This document answers a different question: what should the Interactive Dashboard deliverable actually show, and why — so the team can build it directly from a shared plan instead of everyone guessing at chart choices independently.

The visualization problem we're solving: leadership needs to look at any segment of the business — a region, category, seller, or time period — and immediately see whether it's at risk, and why. The dashboard's job is not to repeat the report; it's to let someone explore a segment they haven't asked about yet.

2. Cross-Validation With Independent Team Analysis

A teammate independently built their own data cleaning and EDA scripts (01_data_cleaning.py, 02_exploratory_analysis.py, insights_summary.md) without reference to our master_orders_enriched.csv pipeline. This is a valuable check — two independent approaches reached the same core conclusions:

- Late delivery is the strongest driver of dissatisfaction
- Deliveries past ~25 days show a sharp jump in 1-star reviews
- Multi-seller orders score significantly worse
- Same-state seller/customer pairs perform better
- Category matters less than logistics factors

This independent replication is good evidence our headline findings are robust, not an artifact of one team member's specific cleaning choices.

That said, there are methodology differences worth resolving before the whole team builds on one dataset:

Gap in the independent version	Why it matters
No dedup of order_reviews (551 orders have duplicate review rows)	Can silently inflate order counts if merged directly instead of aggregated
No aggregation of order_payments (some orders have multiple payment rows)	Same risk of duplicate/inflated rows on merge
product_category_name not translated to English	Category charts will show Portuguese names — fine internally, but not presentation-ready
No documented missing-value or outlier check	Can't confirm delivery-time outliers were verified as real vs. data errors

Recommendation: standardize on ONE canonical dataset for the dashboard — master_orders_enriched.csv (98,666 rows x 40 columns) — so every chart in the dashboard uses consistent, documented cleaning logic. This avoids the team presenting slightly different numbers for the same metric.

3. Recommended Dashboard Structure

Global filters (apply across every page/chart, this is what makes it 'interactive' per the brief rather than a set of static charts):

- Date range (order purchase date)
- Customer state
- Product category
- Delivery outcome (on-time / late)

Suggested page/tab structure — four focused views rather than one crowded page:

Page	Answers
1. Overview	Is the business healthy right now, and is it trending better or worse?
2. Delivery & Logistics	Where and why are deliveries failing?
3. Seller Risk	Which sellers are the actual problem, and why (speed vs. quality)?
4. Category & Geography	Which categories/regions need attention?

4. Chart-by-Chart Specification

Each chart below lists: type, what it's built from, what question it answers, and priority. Must-have charts directly support the report's core claims; Stretch charts are valuable additions if time allows.

Page 1 — Overview

Chart	Type	Data / fields	Question answered	Priority
KPI cards	4 number cards	n_orders, avg review_score, % is_late, avg delivery_days	Is the business healthy right now?	Must-have
Order volume trend	Line chart	purchase_month, count of orders	Is the business growing?	Must-have
Late-rate trend	Line chart (overlaid or below volume)	purchase_month, is_late rate	Is service quality trending up or down as we grow?	Must-have

Page 2 — Delivery & Logistics

Chart	Type	Data / fields	Question answered	Priority
Delivery-time cliff	Bar chart	delivery_days bucketed (0-5,6-10,11-15,16-25,25+) vs avg review_score	Where does satisfaction collapse?	Must-have
Delivery stage breakdown	Grouped/stacked bar	approval_lag_hours, seller_handling_days, carrier_transit_days vs review_score	Which stage of fulfillment to fix?	Must-have
Distance vs delivery time & score	Dual-axis bar+line	distance_km bucketed vs avg delivery_days (bars) and avg review_score (line)	Does distance matter directly, or only through delay?	Stretch
Multi-seller impact	Bar chart	n_sellers vs avg review_score	Does order fragmentation hurt satisfaction?	Must-have

Page 3 — Seller Risk

Chart	Type	Data / fields	Question answered	Priority
Seller scorecard table	Sortable/filterable data table	seller_id, seller_n_orders, seller_revenue, seller_avg_score, seller_late_rate	Which specific sellers need intervention?	Must-have
Revenue concentration	Pareto / cumulative line chart	sellers ranked by revenue, cumulative % of total	How much does the business depend on a few sellers?	Must-have
Speed vs quality quadrant	Scatter plot	x = seller_late_rate, y = seller_avg_score, size = seller_n_orders	Is a seller's problem shipping speed or product quality?	Stretch — directly operationalizes the '\$188K/3.34-star' finding

Page 4 — Category & Geography

Chart	Type	Data / fields	Question answered	Priority
State map	Choropleth map	customer_state, avg review_score or late_rate (toggle)	Which regions need logistics investment?	Must-have
Category comparison	Horizontal bar, sorted	main_category vs avg review_score (min. 200 orders)	Which product categories underperform?	Must-have
Comment behavior	Bar chart	review_score vs % of reviews with a written comment	Where is qualitative complaint data concentrated?	Stretch

5. Priority Ranking: Must-Use vs. Should-Use Charts

Section 4 lists every candidate chart per dashboard page. This section ranks them across the whole dashboard, with the specific evidence that justifies each ranking — so the team can build in the right order if time runs short.

5.1 Must-use — these directly carry the core findings

#	Chart	Type	Why it's non-negotiable
1	Delivery-time cliff	Bar: delivery bucket vs avg review_score	Single strongest finding ($r = -0.30$, 43.6% one-star past 25 days). If the dashboard had only one chart, it would be this one — it directly proves the headline claim of the whole project.
2	Multi-seller impact	Bar: n_sellers vs avg review_score	Most surprising, non-obvious finding (4.12 $\rightarrow$ 2.88 $\rightarrow$ 2.35). Non-obvious findings are what make a project memorable to leadership — 'late = bad' doesn't need a dashboard to explain it, but this one does.
3	Seller scorecard table	Sortable/filterable data table	The only chart that lets leadership actually ACT — drill into a specific seller_id and decide to intervene. Every other chart informs; this one enables a decision, which is literally what an interactive dashboard is for.
4	State map	Choropleth: avg score or late_rate by state	The brief explicitly asks 'which regions' — a table of state names doesn't answer that as fast as a map. Nothing conveys geographic concentration of risk as instantly.
5	Delivery stage breakdown	Grouped bar: approval lag / seller handling / carrier transit vs score	Without this, 'fix delivery' is vague advice. With it, we can say precisely WHICH stage to fix (carrier transit, $r = -0.30$, vs seller handling, $r = -0.16$) — turns a diagnosis into an actionable recommendation.

5.2 Should-use — strong value-add, not load-bearing

#	Chart	Type	Reason
6	Revenue concentration (Pareto)	Cumulative % line	Justifies WHY seller-level risk matters at all — top 1% of sellers generate 25.7% of revenue. Without this, 'watch your sellers' sounds generic; with it, it's clearly high-stakes.
7	Category comparison	Sorted horizontal bar	The brief explicitly asks 'which categories' — cheap to build, directly answers a named question in the problem statement, even though the effect size is smaller than logistics factors.
8	Order volume + late-rate trend	Dual-line chart over time	Gives forward-looking context nothing else provides — is the business improving or getting worse as it scales?
9	Speed vs. quality quadrant	Scatter: late_rate (x) vs avg_score (y), sized by order volume	Visually proves the sharpest single insight — the \$188K seller with a LOW late rate but a terrible score. A scatter shows this instantly; a table row would bury it.

5.3 Deliberately excluded — and why

- Freight / payment-type charts — already validated as near-zero effect ($r = -0.09$, flat across payment types). Charting a null result wastes dashboard space and viewer attention on the weakest evidence.
- Comment-length / comment-rate chart — interesting, but about FUTURE work (text mining), not a current risk signal leadership can act on today. Fine as a report footnote, not dashboard real estate.

Rule of thumb for defending this list if challenged: every must-use chart maps to a correlation or effect size already validated at $r \geq 0.15$, or a >1 -star average score gap. The should-use charts answer questions the brief asks explicitly, even where the effect is smaller.

6. Tooling Options

Tool	Good fit if...	Tradeoff
Streamlit (Python)	Team is comfortable in Python, wants fast build, needs filters + tables + charts	Less polished default styling than BI tools
Plotly Dash (Python)	Team wants more custom interactivity (e.g. the quadrant scatter, drill-down)	Steeper learning curve than Streamlit
Tableau / Power BI	Team wants drag-and-drop, no-code, polished maps out of the box	Less flexible for custom logic (e.g. the 3-stage delivery decomposition)

Given the team already has Python scripts (pandas-based cleaning/EDA), Streamlit is the lowest-friction choice — it can load `master_orders_enriched.csv` directly and most of these charts are a few lines each with `plotly/matplotlib` inside it.

7. Next Steps

- Agree as a team to standardize on `master_orders_enriched.csv` as the single source of truth (Section 2).
- Assign each dashboard page (Section 3-4) to a team member.
- Build Must-have charts first; treat Stretch charts as time allows.
- Once built, sanity-check each chart's numbers against the figures already verified in `Team_Report_EDA_Findings.docx` — if they don't match, the dashboard has a bug, not a new finding.
- Update individual contribution logs as each page is built.